

Seongmin Jung

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RESEARCH INTEREST

Multimodal Robot Learning, Vision-Language(-Action) Models, 3D Computer Vision, Embodied AI

EDUCATION

Seoul National University

M.S. in Artificial Intelligence (Advisor: Prof. Jongwoo Lim)

GPA: 4.25/4.30

Sep 2024 – Aug 2026 (Expected)

Seoul, South Korea

Yonsei University

B.S. in Electrical and Electronic Engineering

GPA: 4.02/4.30 • Graduated with **High Honors (Top 3% in college)**

Mar 2019 – Feb 2024

Seoul, South Korea

Case Western Reserve University

Exchange Program

Aug 2022 – May 2023

Cleveland, OH, USA

HONORS AND AWARDS

The National Scholarship for Science and Engineering

Korea Student Aid Foundation (KOSAF)

- Full merit-based scholarship awarded to top STEM students by the South Korean government, covering full tuition.

2019 – 2024

Seoul, South Korea

High Honors

Dept. of Electrical & Electronic Engineering, Yonsei University

Fall 2019

Seoul, South Korea

Highest Honors

Dept. of Electrical & Electronic Engineering, Yonsei University

Spring 2019

Seoul, South Korea

PREPRINTS & MANUSCRIPTS

- [1] **Seongmin Jung***, Seongho Choi*, Gunwoo Jeon, Minsu Cho, and Jongwoo Lim. PanoGrounder: Bridging 2D and 3D with Panoramic Scene Representations for VLM-based 3D Visual Grounding. *ECCV 2026, under review*.

ArXiv: <https://arxiv.org/abs/2512.20907>

RESEARCH EXPERIENCES

AI4CE Lab, New York University

Internship (Advisor: Prof. Chen Feng)

Tactile-Informed Imitation Learning

Remote / New York City, NY, USA

Nov 2025 – Present

- Developing visual-tactile compositional diffusion policies that fuse low-rate visual inputs with high-rate piezoelectric tactile signals for contact-rich manipulation.
- Constructed a LeRobot-based training pipeline, reimplementing baselines for integration with the custom robot hardware.

Robot Vision Lab, Seoul National University

Graduate Research

Solving Rubik's Cube with Bimanual Robot Hands

Seoul, South Korea

Sep 2025 - Present

- Collected VR teleoperation data on a Fourier GR-1 bimanual humanoid robot in Isaac Lab using a Meta Quest 3.
- Fine-tuned an open-source VLA (GR00T N1.5) for stable grasping and rotation of Rubik's Cube faces.

PanoGrounder

Jan 2025 – Mar 2026

- Proposed a framework for VLMs reasoning over 3D Mesh/3DGS via panoramic rendering, enabling spatial understanding.
- Fine-tuned a ~17B VLM on 8x A100 GPUs using DeepSpeed/LoRA, injecting DINOv2 features via custom adapters.
- Achieved SOTA (+4.7% on Nr3D), robust generalization (+17.4% on ARKitScenes), and efficient inference (~3.3s/query).

Neural SLAM with Point Cloud Projection for Faster RGB-D Tracking

Sep 2024 - Dec 2024

- Proposed a point-NeRF-based differentiable camera pose tracker, replacing iterative NeRF queries along camera rays.
- Replaced the tracking module of Point-SLAM, achieving over 3× faster tracking while preserving trajectory accuracy.

Biological Cybernetics Lab, Yonsei University

Undergraduate Thesis (Advisor: Prof. DaeEun Kim)

Autonomous Navigation for High-Speed Ground Robot

Seoul, South Korea

Jul 2023 – Dec 2023

- Implemented low-latency obstacle avoidance and path planning in C++, using ROS and LiDAR odometry.
- Deployed on a Raspberry-Pi-powered RC car achieving high-speed navigation at max 5m/s.

PROFESSIONAL EXPERIENCE

Stealth Mode MedTech Startup

3D Vision Engineer

Oct 2025 – Present
Remote / Pittsburgh, PA, USA

- Developing 3D volumetric segmentation networks (e.g., 3D U-Net) for processing noisy, high-resolution voxel data.
- Building data workflow, 5-fold cross-validation, and clinical evaluation.
- Collaborating with medical professionals to define clinical problems and enhance performance for real-world deployment.

Mobiltech

Research Assistant

Aug 2023 – Jun 2024
Seoul, South Korea

Enhancing Pose Tracking in Urban Scenes

- Developed a LiDAR odometry pipeline in C++, using per-frame height maps of surrounding structures, improving trajectory accuracy and robustness in dense urban scenes.

Isaac Sim Environment for Heterogeneous Multi-Robot Systems

- Implemented a heterogeneous multi-robot simulation with ground and aerial platforms in Isaac Sim, integrating per-robot control and sensor setups in ROS.

TEACHING EXPERIENCES

Digital Computer Concept and Practice (F37.202), *Seoul National University*

Teaching Assistant

Spring 2025, Spring 2026
Seoul, South Korea

- Led weekly C++ programming lab sessions and developed course materials and assignments.

PATENTS

- [1] Y. Lee, **S. Jung**, J. Jo, and Y. Kim. 2023. Apparatus for collecting cigarette butts with IoT. Korea Patent 1028688440000, September 30, 2025.

CLUB ACTIVITIES

CWRUbotix, *Robotics Club at Case Western Reserve University*

MATE ROV World Championship 2023

Dec 2022 – Jun 2023
Cleveland, OH, USA

- Constructed underwater robot simulation for mission environment using ROS and Gazebo Simulation.
- Designed and implemented an Arduino-based wireless communication module for mission execution.

VERY, *Entrepreneurship Club at Yonsei University*

Team Lead

Mar 2021 – Feb 2022
Seoul, South Korea

- Established online music ensemble platform based on self-recording for college bands.
- Led a 4-member team, coordinating operation, design, and development throughout the project.

LEADERSHIP

Korean Mentor

Office of International Affairs, Yonsei University

Sep 2023 – Aug 2024
Seoul, South Korea

- Mentored 10 incoming study-abroad students, aiding in cultural and academic adaptation in Korea.

President of Student Council

Dept. of Electrical & Electrical Engineering, Yonsei University

Nov 2020 – Oct 2021
Seoul, South Korea

- Led key events including freshmen orientation and homecoming, organizing 300+ students.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, MATLAB, Verilog

Libraries & Frameworks: PyTorch, DeepSpeed, LeRobot, OpenCV, Open3D, 3DGS, NeRF

Robotics & Simulation: ROS 1 & 2, Isaac Sim, Isaac Lab, Gazebo, VR Teleoperation (Meta Quest3 + XR Robotics)

Languages: Korean (native), English (fluent)

- TOEFL iBT: 108 (Reading: 30, Listening: 29, Speaking: 24, Writing: 25)